BASE24 Interac

KEK Change Procedure Using TSS

Version 2.1

May 2008

ACI Worldwide (Canada) Inc.
200 Wellington St. W. Suite 700
Toronto, Ontario, M5V 3C7
(416) 813-3000

Revision History

Record of Changes

Author	Date	Version	Comments
IC	March 2005	1.0	Initial version
JLF	October 2005	1.1	Internal review
DN	March 2007	1.2	Change document title & update section Introduction to reflect TSS 4.4 & higher
MM	April 2007	1.3	Add steps to quiesce and activate the IMN connection
MM	March 2008	2.0	To incorporate suggested process from Carlo Tuzi & add clarification
DN	May 2008	2.1	Change document title & add warning note in sub-section 2.3 re. adding new ISEC record

Document Reference

Document Name	Author	Version/Date
Transaction Security Services Processing Guide	ACI	Release 3.3 Oct 2003
ACI Desktop Manual	ACI	

Table of Contents

ACI Worldwide (Canada) Inc.

1. INTRODUCTION

This document describes the key exchange keys (KEK) change procedure applicable to TSS 4.4. or higher.

The basic steps which are described in this document cover two main areas:

1. The preparation of the environment in order to perform the KEK change, and
2. The actual steps required to implement the new KEK values.

1.1 Preparation Activities

1. Add New Interface Security Configuration records (ISEC) for each Connector.
2. ISEC records need to be added for the Acquirer and Issuer.
3. Modify the new ISEC records for each connector replacing the existing KEK cryptograms with the new KEK cryptogram, for all appropriate KEY tabs.
4. Issue TSS Commands to rebuild extended memory Tables and warmboot the ISLO processes after all new KEK have been entered.

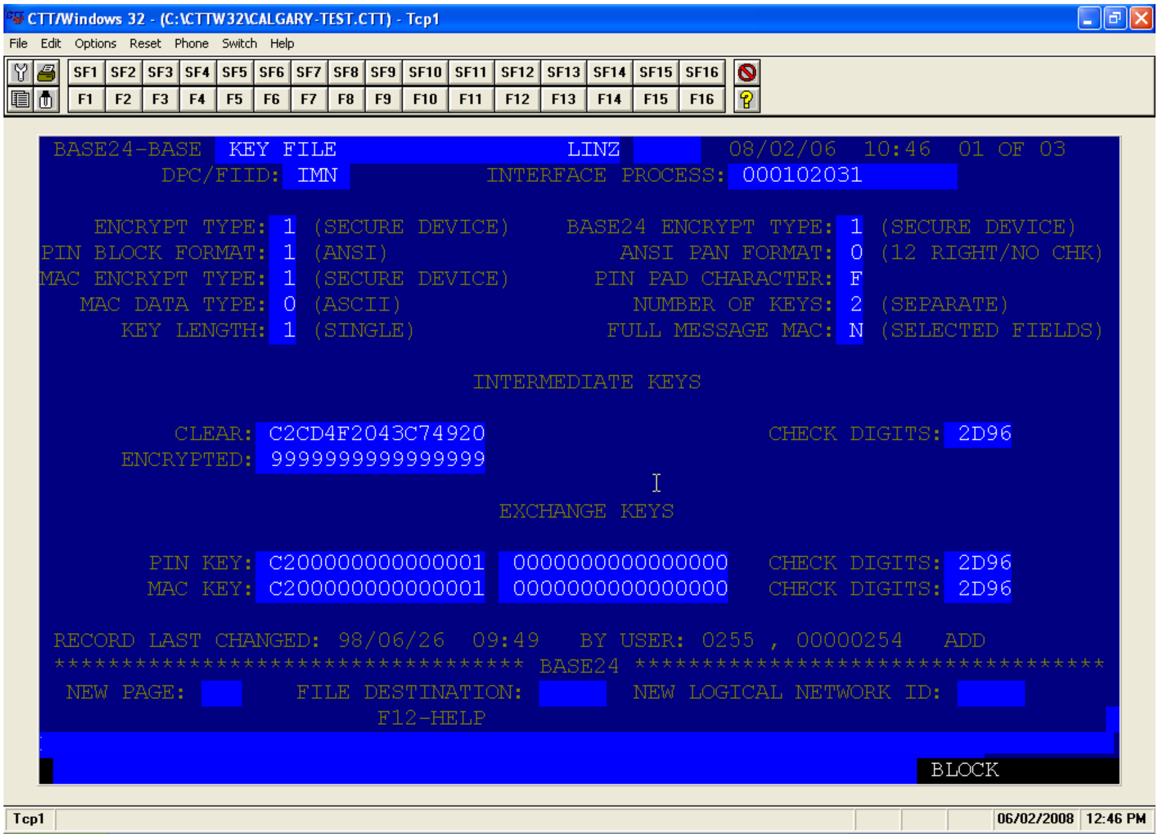
1.2 Installation Activities (Implementation Morning)

1. Change each KEYF in BASE24 to make reference to the new Key tags for each connector.
2. Changes are required in multiple locations in the KEYF (screen 1 and 2).
3. Issue Warmboot commands to all IMNi or IMNsc processes.

2. PREPARATION EXAMPLE

The following is an example of a BASE24 KEYF record for a connector's issuer record.

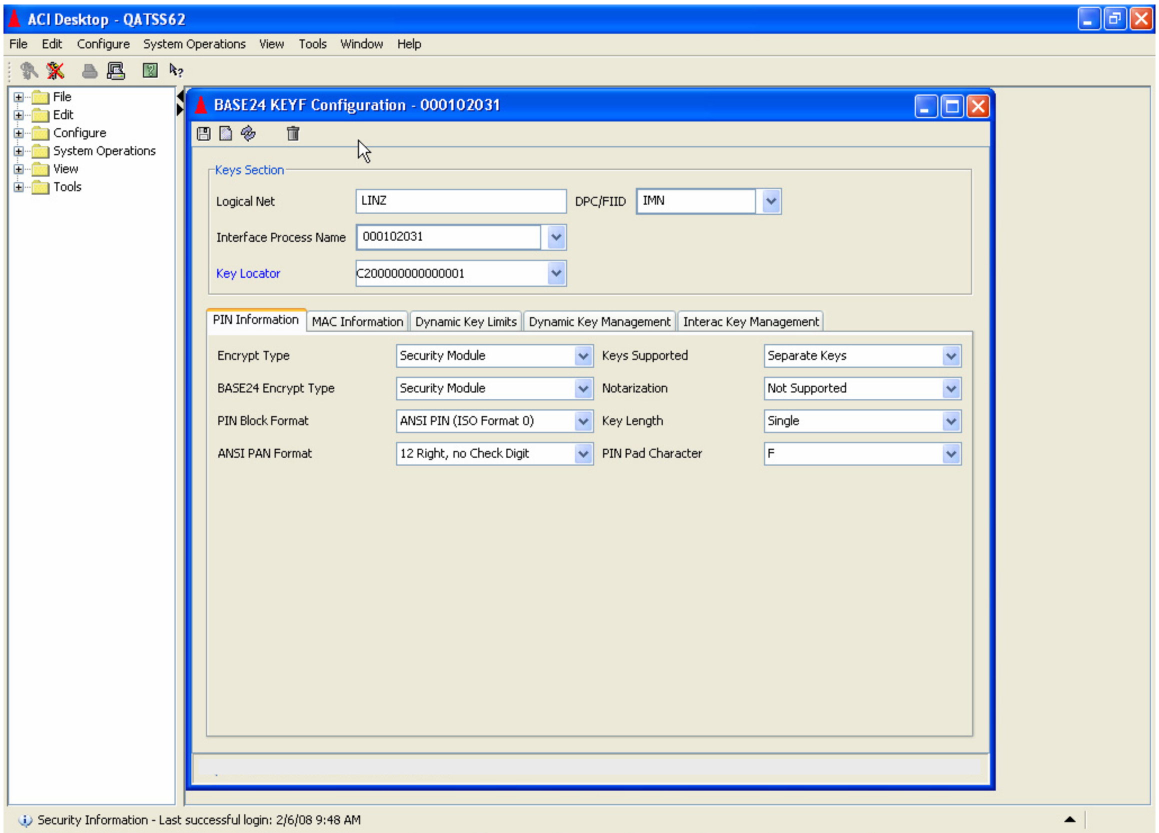
It shows that the current tag in use for this record is "C200000000000001".



2.1 Step 1: Read existing TSS KEYF record

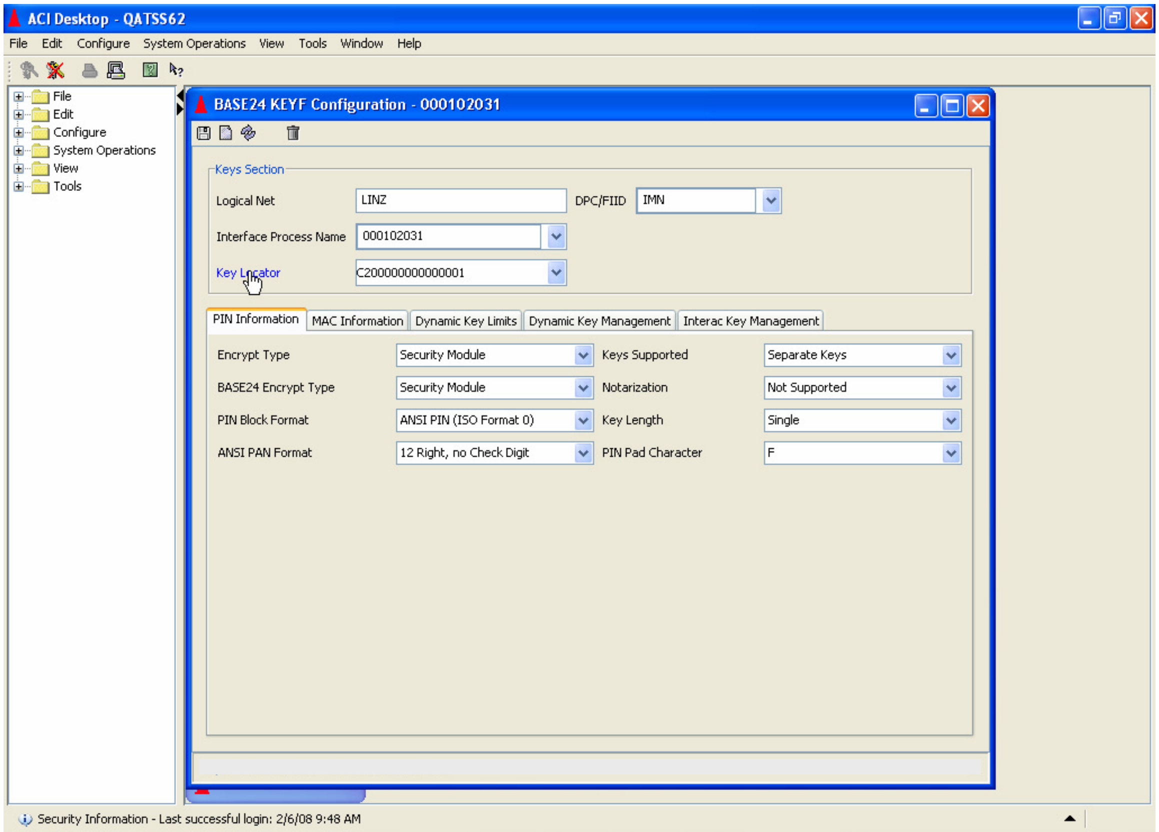
Read the TSS KEYF interface record for the connector that the new Key Locator will be added for.

In this example, this is the connector's issuer record as contained within the BASE24 KEYF record.



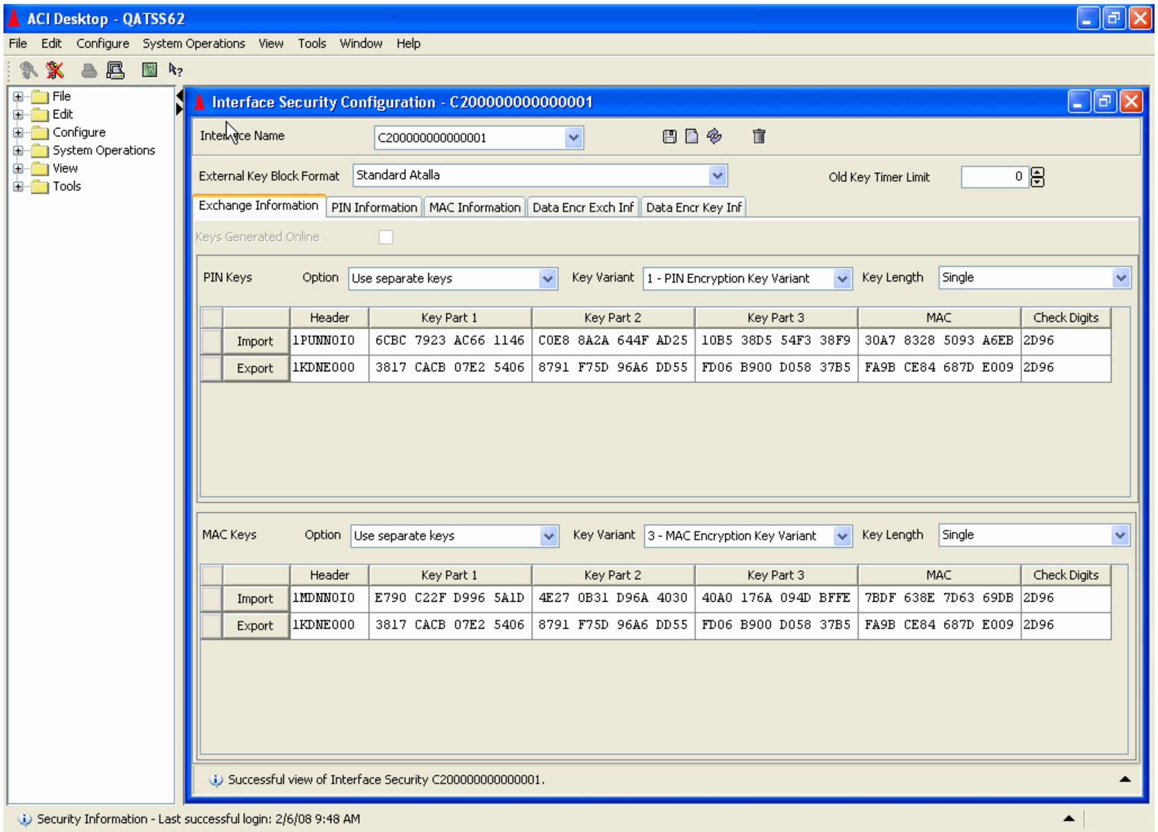
2.2 Step 2: Read existing KEYF record from TSS

Select Key Locator in order to bring up Interchange Security Configuration record for this connector.



2.3 Step 3: Interface Security Configuration (ISEC) record

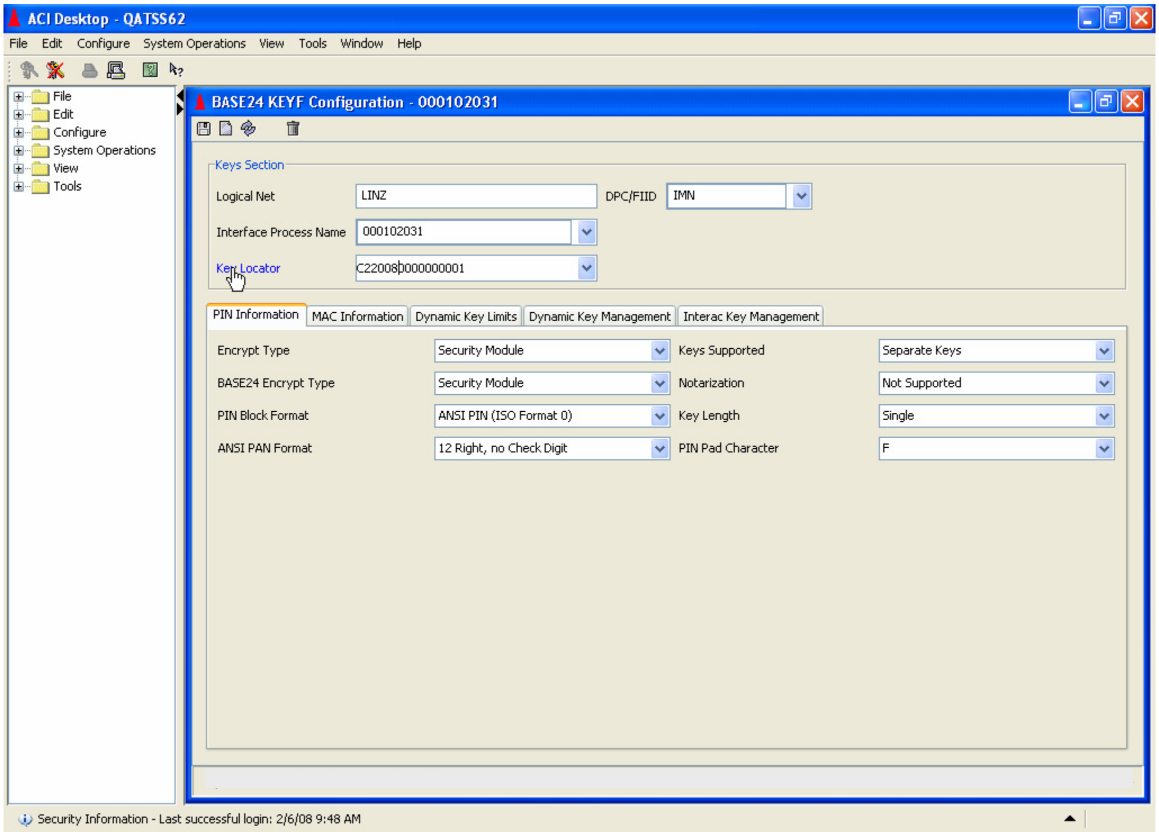
The ISEC record will be now be displayed.



- Minimize the TSS ISEC record read.
- Expand the TSS KEYF record.

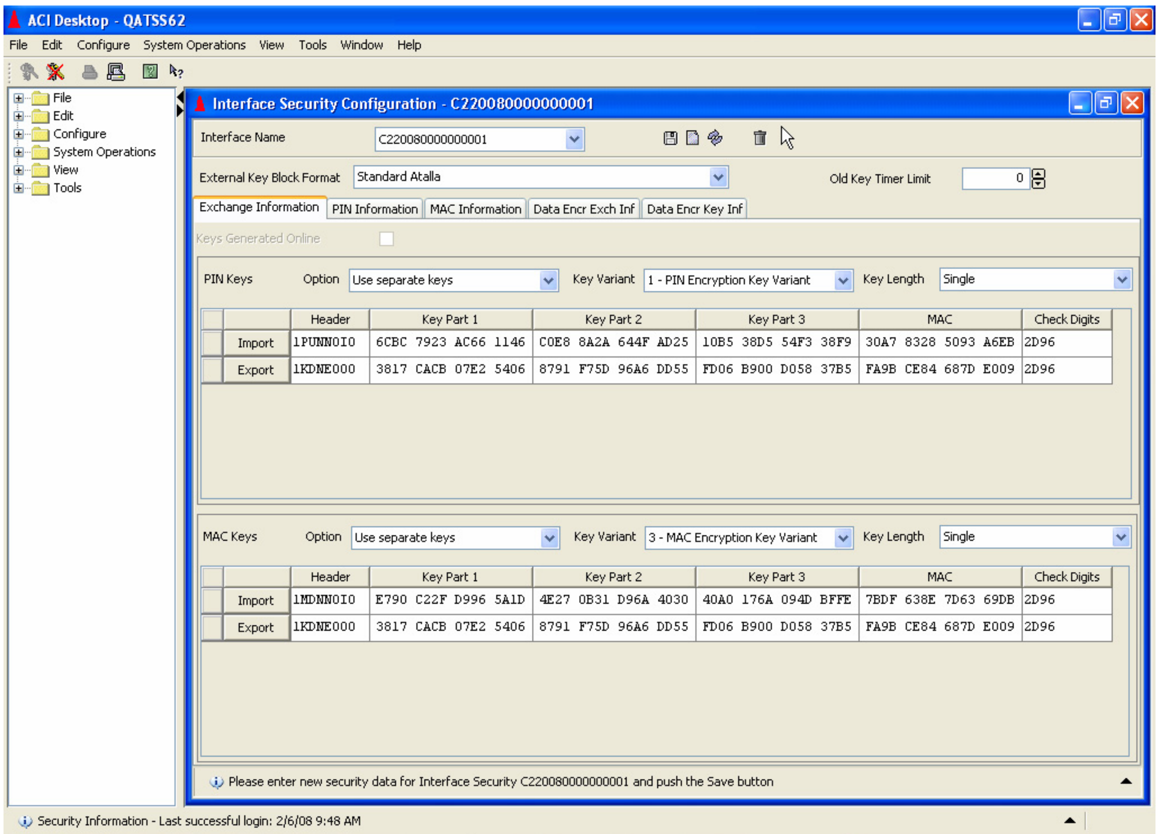
- Change the current Key locator to a different unique value.
  In this example C20000000000001 is changed to C22008000000001.

- Select key locator.

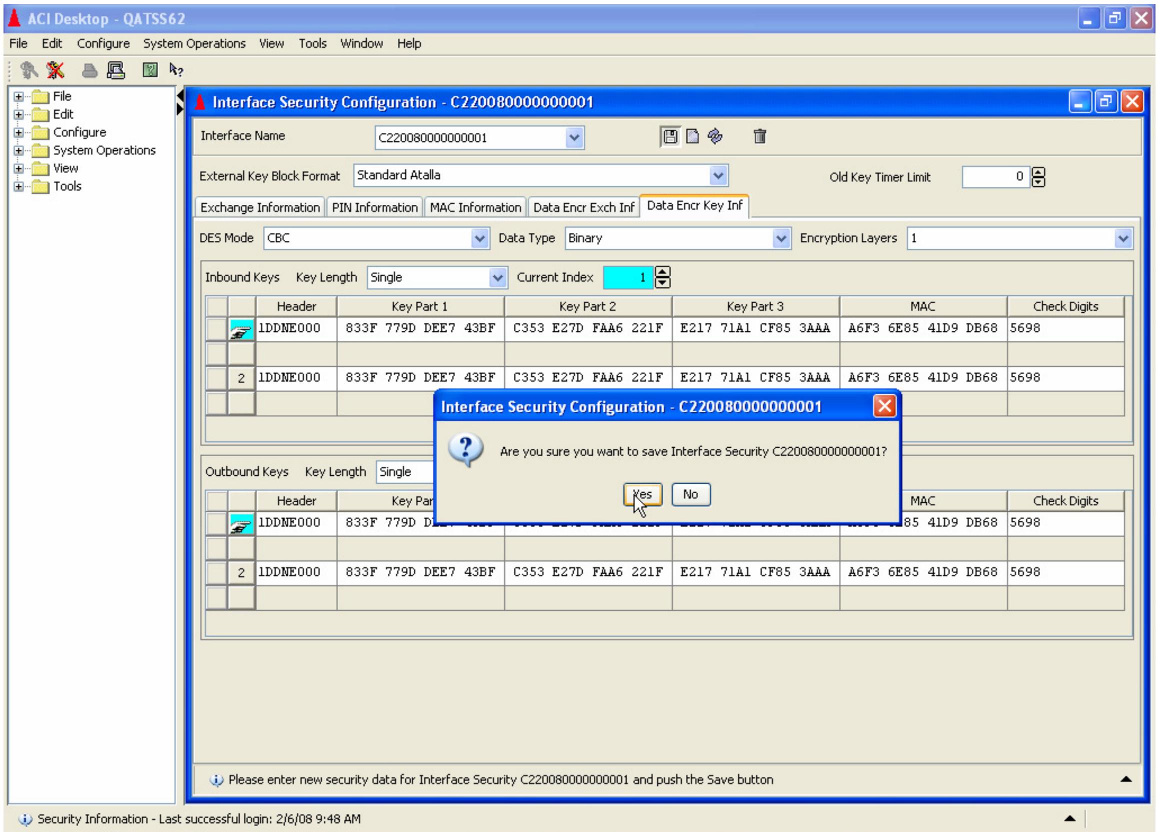


- As a result of selecting the Key locator, a new record with the exact same information and the originally read record will be shown.

  Note that new versions of the ACI Desktop (after version 6.4) may not display the new record with the exact same information as the record originally read. If it does not, the information in the new record must be added using the contents of the old record.

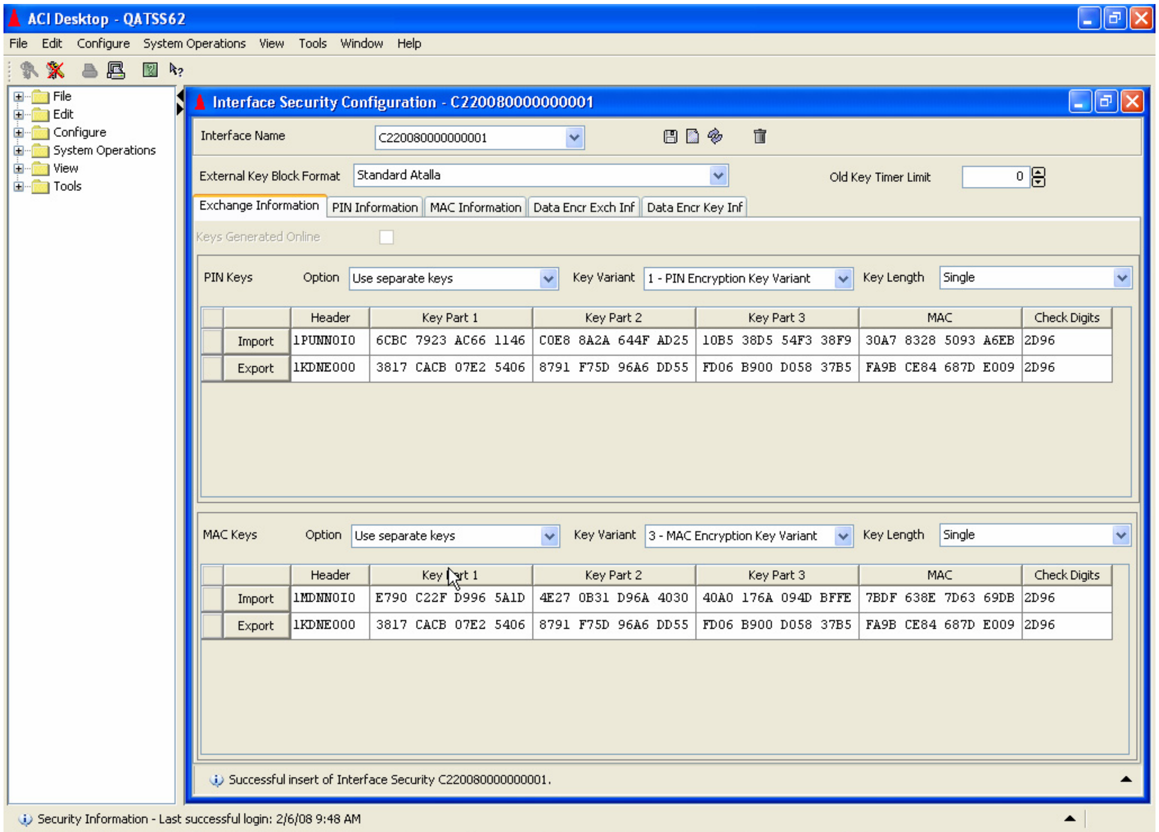


- Validate that the key locator value is what you expect it to be.
- Add the records using the "**diskette Icon**".



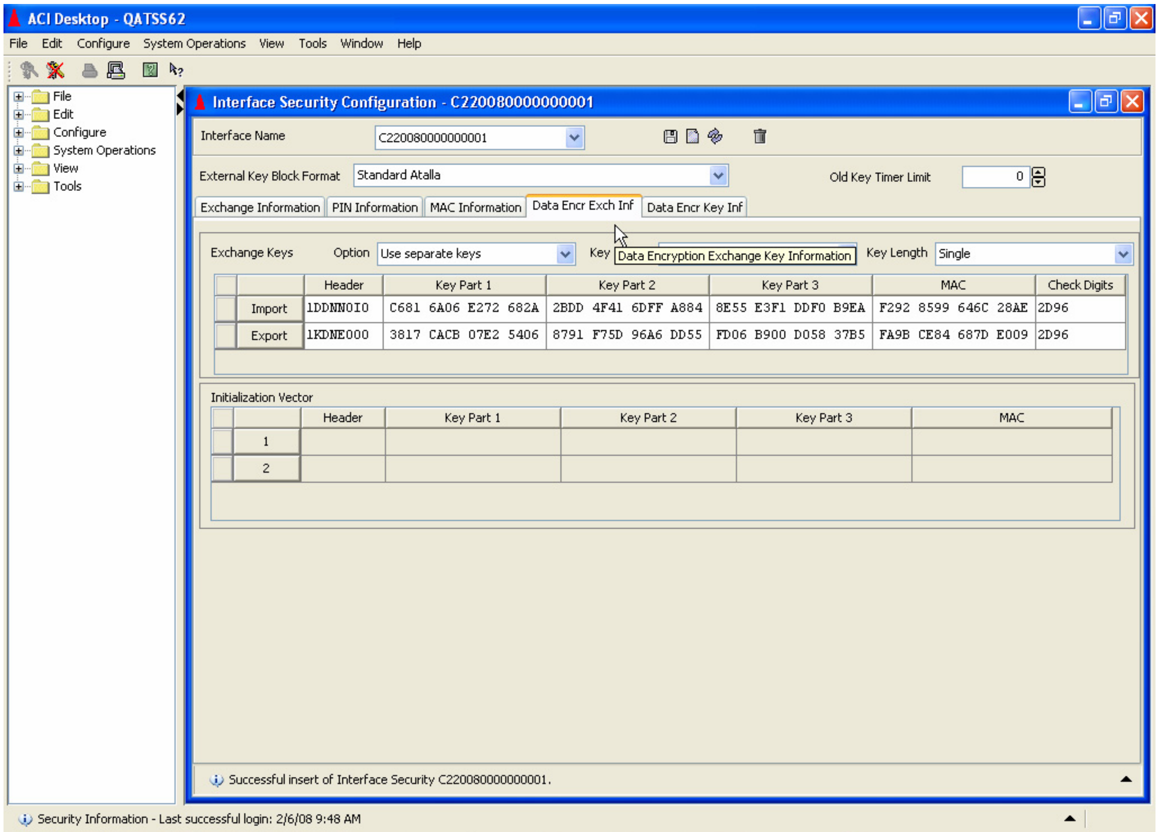
- Confirm add of the record.

2.4 Step 4: Updating of New ISEC record



- Validate that the Interface security record is added.
- At this point, the new ISEC record has been added, and you may proceed to modify the KEK values in the appropriate KEY tabs.
  - Re-read the record.
  - Select the "EXCHANGE INFORMATION" -TAB, and update the various keys.

For IDP Only, the Data Encryption Exchange Key Information values also need to be updated.



- Save the changed record, using the **"diskette Icon"**.

  - The above is repeated for every connector for the Acquirer and Issuer key locators.

  - Once all of the records have been added and updated, issue the following commands in NCPCOM to re-build the extended memory table for the ISEC records and warmboot ISLO process.

    The first command to the process will re-build the extended memory for the ISEC table. This command must be sent only to one ISLO process, if multiple ISLO processes are available. In this example, the ISLO process is located in XPNET Node C.

    The second and third commands must be sent to ALL ISLO processes.

- deliver process P1C^ISLO, "DELIVER DSLDR INTERCHANGE_SECURITY_OLTP"
- deliver process P1*^ISLO*, "ALTER DASRC INTERCHANGE_SECURITY_OLTP"
- deliver process P1*^ISLO*, "ALTER DASRC ICHG_SECURITY"

The following logs should be displayed in EMS log:

```
05-03-24;22:46:33.494 \ACCIS1.$QFIS     ACI.1033.1000             60
          Mgr PPD: \ACCIS1.$QFCN  Generator: P1C^ISLO
          DSLDR: Data source INTERCHANGE_SECURITY_OLTP successfully loaded.
05-03-24;22:46:33.623 \ACCIS1.$QFIS     ACI.1049.1000             60
          Mgr PPD: \ACCIS1.$QFCN  Generator: P1C^ISLO
          DASRC: Successful warmboot of data source:
INTERCHANGE_SECURITY_OLTP
05-03-24;22:46:33.691 \ACCIS1.$QFIS     ACI.1049.1000             60
          Mgr PPD: \ACCIS1.$QFCN  Generator: P1C^ISLO
          DASRC: Successful warmboot of data source: ICHG_SECURITY
```

- Validate the old ISEC memory table is no longer in use by TSS.

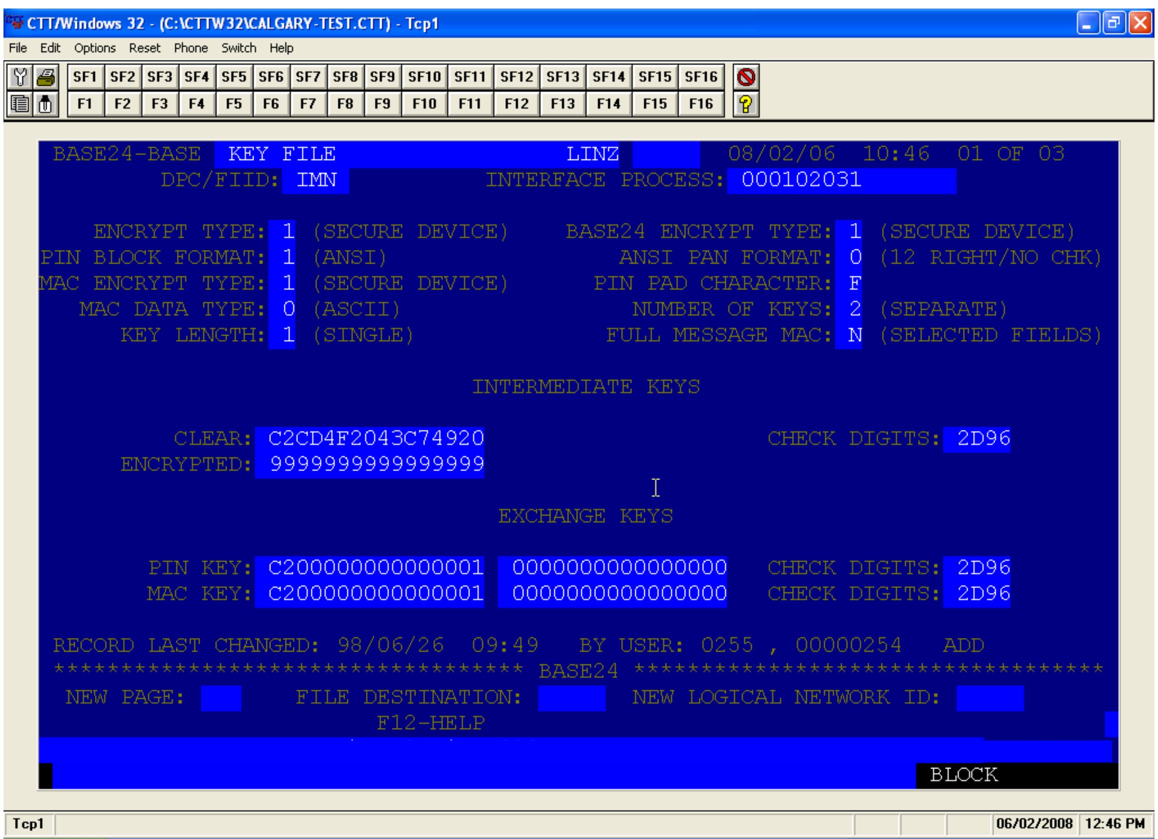
<u>NOTE</u>: **DO NOT save the TSS KEYF RECORD with the modified KEY Locator Value. Simply close it after it is no longer required.**

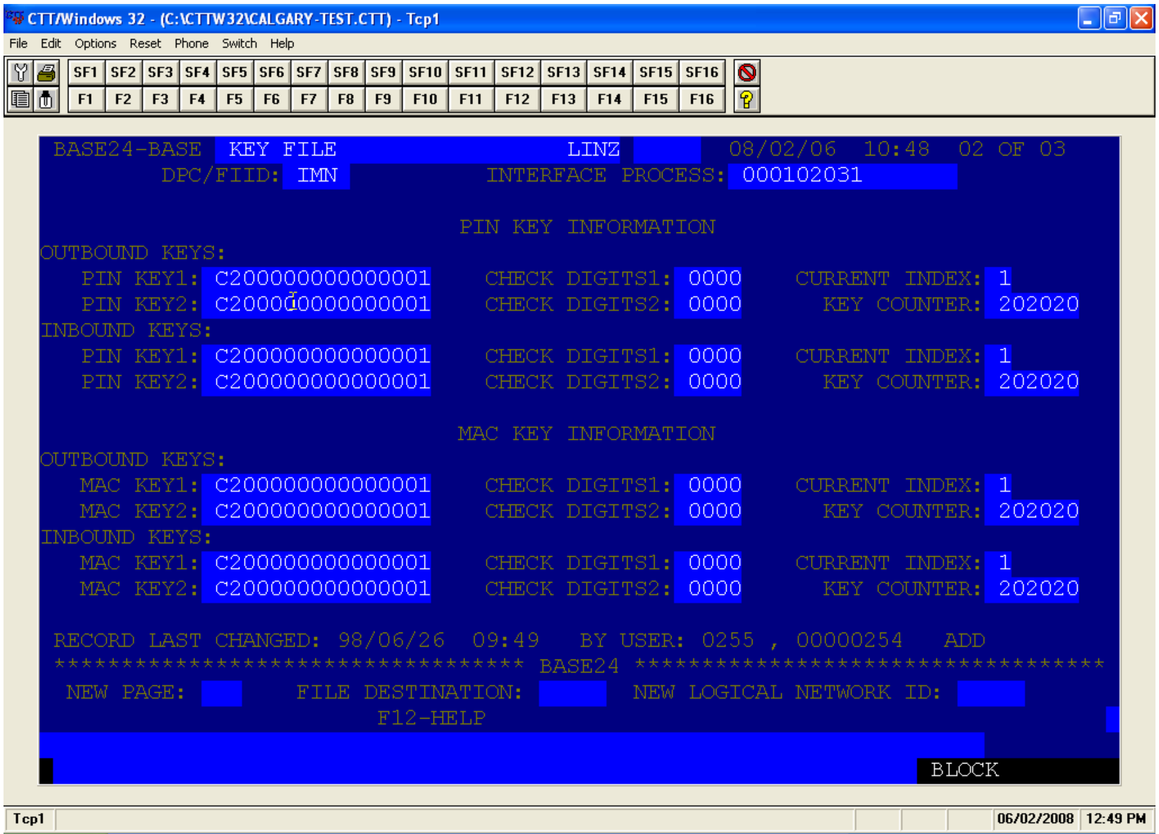
3. INSTALLATION ACTIVITIES (IMPLEMENTATION MORNING)

The morning that the keys are to be implemented, the activity required is to modify every BASE24 KEYF record for both the Acquirer and Issuer to change the Key locator values to the new values associated with each connector.

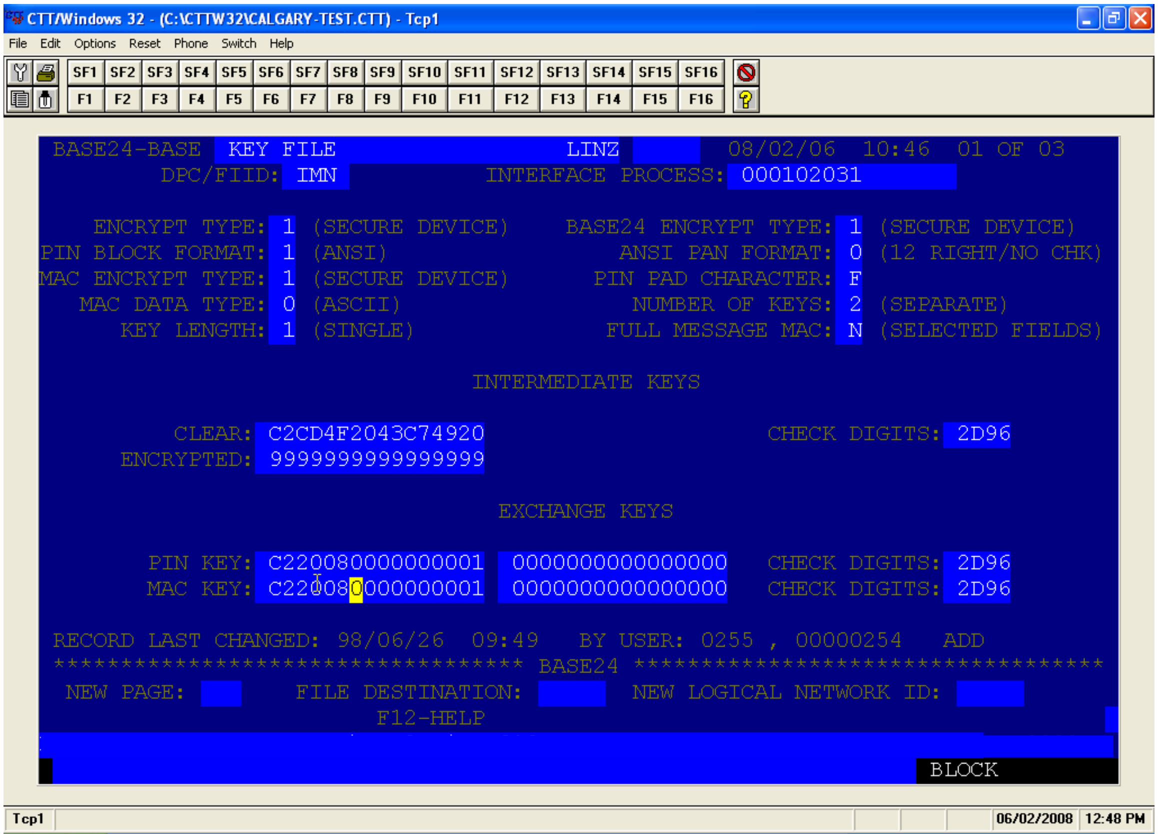
In this example the issuer record is read, which shows the current key locator in use as “C20000000000001” on page 1 and 3. These values will be changed to “C22008000000001”, to match the Key Locator record which was added in the preparation steps.

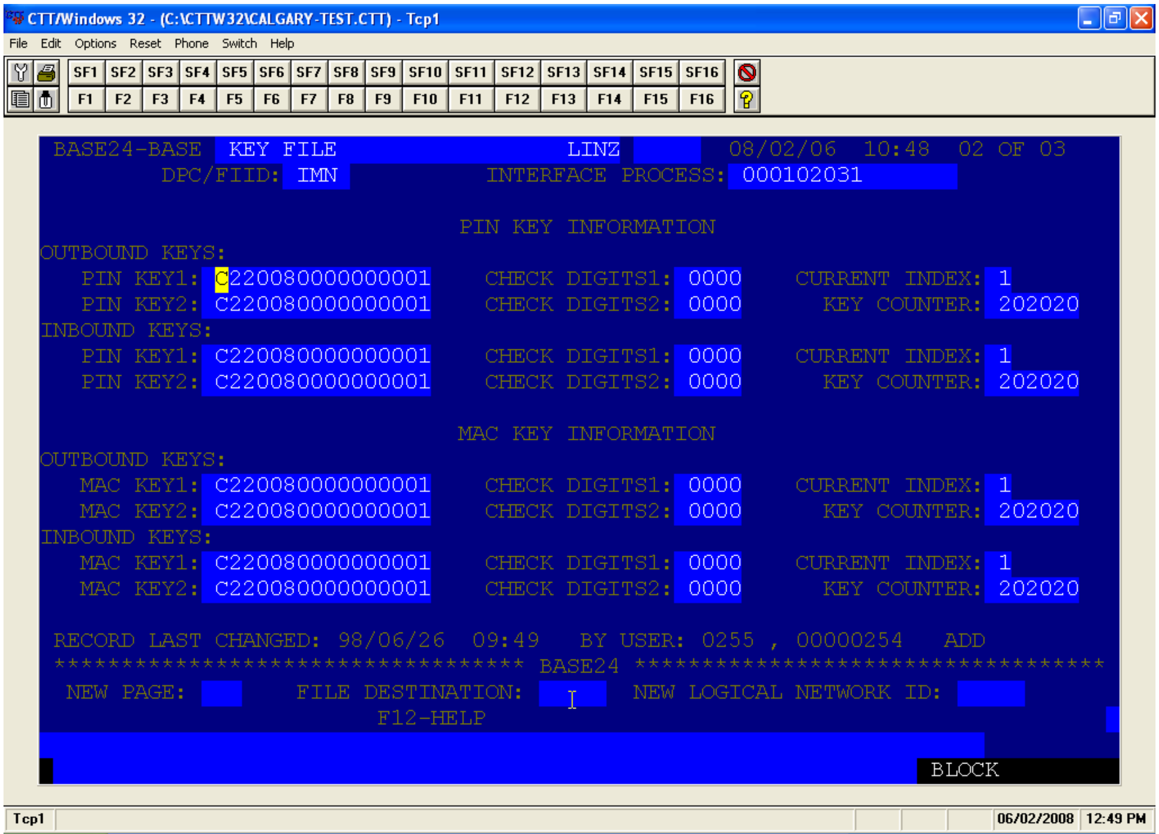
The Check digits of the KEK are also changed in the BASE24 KEYF record for future reference of the KEK check digits.




CTT/Windows 32 - (C:\CTTW32\CALGARY-TEST.CTT) - Tcp1
File Edit Options Reset Phone Switch Help
SF1 SF2 SF3 SF4 SF5 SF6 SF7 SF8 SF9 SF10 SF11 SF12 SF13 SF14 SF15 SF16
F1 F2 F3 F4 F5 F6 F7 F8 F9 F10 F11 F12 F13 F14 F15 F16
BASE24-BASE KEY FILE LINZ 08/02/06 10:48 02 OF 03
DPC/FIID: IMN INTERFACE PROCESS: 000102031
PIN KEY INFORMATION
OUTBOUND KEYS:
PIN KEY1: C200000000000001 CHECK DIGITS1: 0000 CURRENT INDEX: 1
PIN KEY2: C200000000000001 CHECK DIGITS2: 0000 KEY COUNTER: 202020
INBOUND KEYS:
PIN KEY1: C200000000000001 CHECK DIGITS1: 0000 CURRENT INDEX: 1
PIN KEY2: C200000000000001 CHECK DIGITS2: 0000 KEY COUNTER: 202020
MAC KEY INFORMATION
OUTBOUND KEYS:
MAC KEY1: C200000000000001 CHECK DIGITS1: 0000 CURRENT INDEX: 1
MAC KEY2: C200000000000001 CHECK DIGITS2: 0000 KEY COUNTER: 202020
INBOUND KEYS:
MAC KEY1: C200000000000001 CHECK DIGITS1: 0000 CURRENT INDEX: 1
MAC KEY2: C200000000000001 CHECK DIGITS2: 0000 KEY COUNTER: 202020
RECORD LAST CHANGED: 98/06/26 09:49 BY USER: 0255 , 00000254 ADD
********************************** BASE24 **********************************
NEW PAGE: FILE DESTINATION: NEW LOGICAL NETWORK ID:
F12-HELP
BLOCK
Tcp1
06/02/2008 12:49 PM

Example of modified BASE24 KEYF screens showing the new Key Locator values.





- These steps are repeated for each connector Acquirer and Issuer records.

- Once all of the records have been updated, and the other members are ready to proceed with the implementation, the IMNi or IMNsc processes can be warmbooted in order for the new key locators to become effective. Please warmboot the IMNi / IMNsc process one at a time. Ensure each IMNi/IMNsc process warmboots successfully before warmbooting the next IMNi / IMNsc process.

- After each remote node has installed the new KEK, deliver new keys to the remote node: MAC key first, then PIN key For IMNi only, the message key is required. **Ensure each key exchange succeeds before proceeding to the next key.**

  - deliver process p1a^imnm, "newmackey 777761 01"
  - deliver process p1a^imnm, "newpinkey 777761 01"
  - deliver process p1a^imnm, "newmsgkey 777761 01" (for IMNi only)

4. CONSIDERATIONS

The process of changing keys has the possibility of causing a number of error conditions between members since the timing of activity can vary from member to member.

The error conditions which may be experienced are:

1. Working Key change errors

2. Transaction declined due to keys not being synchronized.

End Of Document